

Edge of Space LiDAR Payload for Prairie Ecosystem Analysis

Project Overview

As part of the Iowa State University **LEAF (Learning Ecosystems Across Facilities)** research program, I helped design and deploy a drone-mounted sensing payload that combined **LiDAR** and **Structure-from-Motion (SfM)** imaging to measure vegetation height across prairie ecosystems.

The project's objective was to create accurate 3D models of maintained and unmanaged prairie sites while accounting for natural changes in terrain elevation.

My Contributions

Payload Design

- Designed the payload enclosure in **SolidWorks** to fit within the drone's strict size and weight constraints.
- Optimized the layout of the LiDAR sensor, Raspberry Pi, Arduino, camera, battery, and wiring to create a compact, flight-ready system.
- Worked through multiple design iterations to ensure all components fit securely and were accessible for testing and maintenance.

Flight Planning

- Designed the autonomous flight path to achieve **70% image overlap**, providing sufficient coverage for Structure-from-Motion reconstruction.
- Determined flight spacing and image capture requirements to maximize reconstruction accuracy while remaining within the drone's operational limits.

Software Development

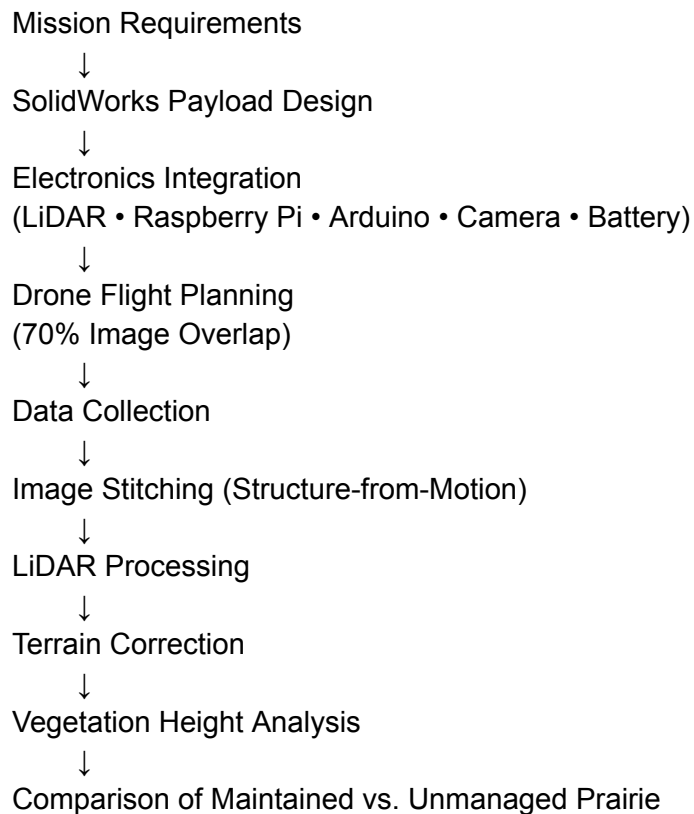
- Assisted in developing the embedded software used for sensor operation and data collection.
- Contributed to the Arduino firmware written in **C** for hardware control and sensor integration.

Data Analysis

- Processed LiDAR and aerial imagery using **R**.
- Stitched overlapping aerial images together to generate high-resolution **3D terrain and vegetation models**.

- Corrected for terrain elevation differences to calculate true plant heights rather than absolute elevation.
- Compared vegetation structure between maintained and unmanaged prairie sites using statistical analysis.

Engineering Workflow



Technologies Used

Mechanical Design

- SolidWorks
- 3D Printing

Embedded Systems

- Arduino (C)
- Raspberry Pi
- Raspberry Pi Camera
- LiDAR

Data Processing

- R
- Structure-from-Motion (SfM)
- Photogrammetry
- LiDAR Point Cloud Analysis

Engineering Skills

- Payload Design
- Systems Integration
- Flight Planning
- Sensor Integration
- Data Analysis
- Scientific Visualization